Training Course

Amazon Web Service

Module 7:

Route 53

Module 7: Route53

- **Goal**: Understanding Route 53 in AWS
- TTL
- CNAME vs Alias
- Health Checks
- Routing Policies
  - ✓ Simple
  - ✓ Weight
  - ✓ Latency
  - ✓ Failover
  - ✓ Geolocation
  - ✓ Multi Value
- 3rd party domains integration

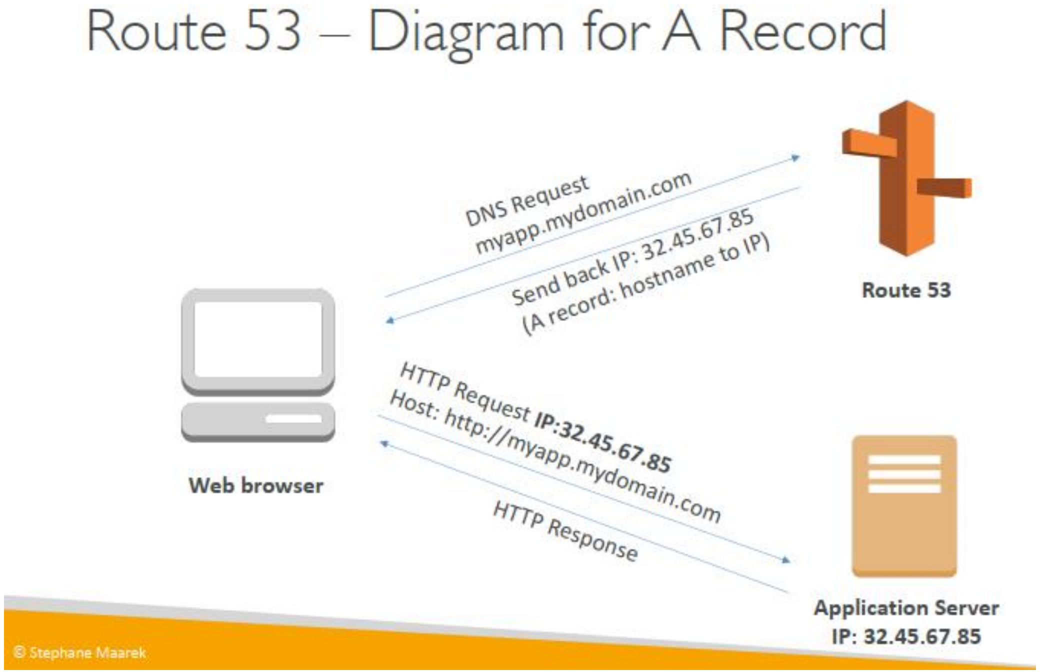
  Lab: Create and configuring Route 53

Section introduction

AWS Route 53 Overview

- Route 53 is a Managed DNS (Domain Name System)
- DNS is a collection of rules and records which helps clients understand how to reach a server through URLs
- In AWS, the most common records are:
  - A: hostname to IPv4
  - AAAA: hostname to IPv6
  - CNAME: hostname to hostname
  - Alias: hostname to AWS resource

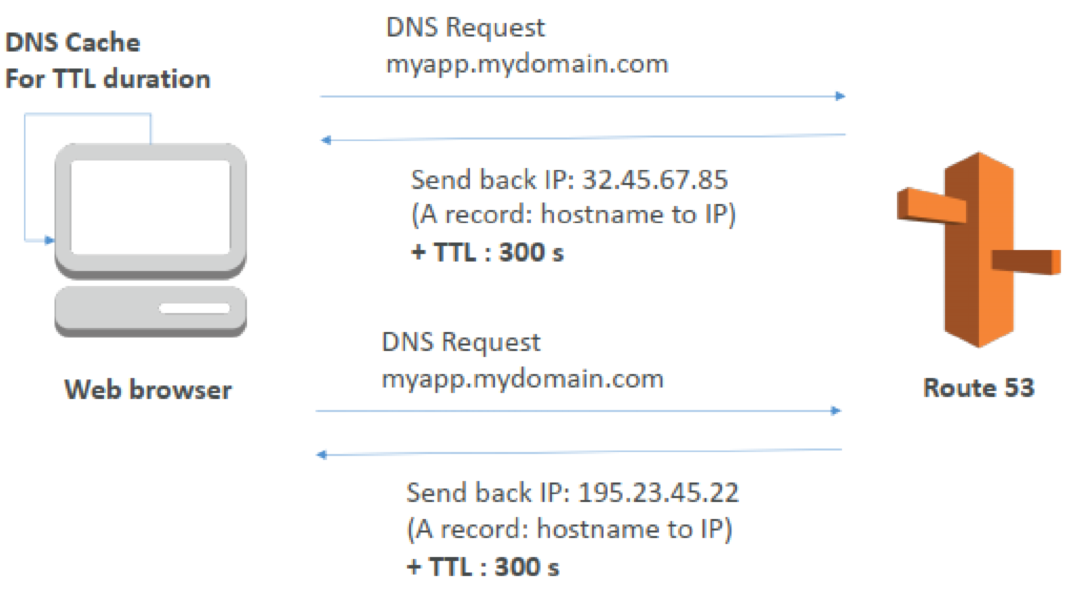
Route 53 – Diagram for A Record



AWS Route 53 Overview

- Route53 can use
  - Public domain names you own (or buy): application1.mypublicdomain.com
  - Private domain names that can be resolved by your instances in your VPCs: application1.company.internal
- Route53 has advanced features such as
  - Load balancing (through DNS – also called client load balancing)
  - Health checks (although limited…)
  - Routing policy: simple, failover, geolocation, latency, weighted, multi value
- You pay $0.50 per month per hosted zone

DNS Records TTL (Time to Live)



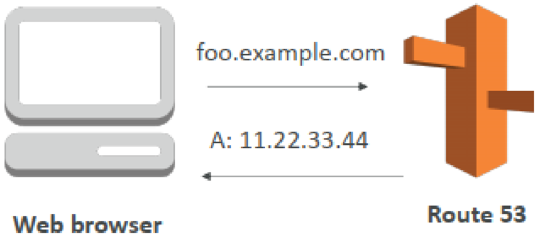
- High TTL: (e.g. 24hr)
  - Less traffic on DNS
  - Possibly outdated records
- Low TTL: (e.g 60 s)
  - More traffic on DNS
  - Records are outdated for less time
  - Easy to change records
- TTL is mandatory for each DNS record

CNAME vs Alias

- AWS Resources (Load Balancer, CloudFront…) expose an AWS hostname: lb1-1234.us-east-2.elb.amazonaws.com and you want myapp.mydomain.com
- CNAME:
  - Points a hostname to any other hostname (app.mydomain.com -> blabla.anything.com)
  - ONLY FOR NON ROOT DOMAIN (aka.somthing.mydomain.com)
- Alias
  - Point a hostname to an AWS Resource (app.mydomain.com -> blabla.amazonaws.com)
  - Work for ROOT DOMAIN and NON ROOT DOMAIN (aka.mydomain.com)
  - Free of charge
  - Native healthy check

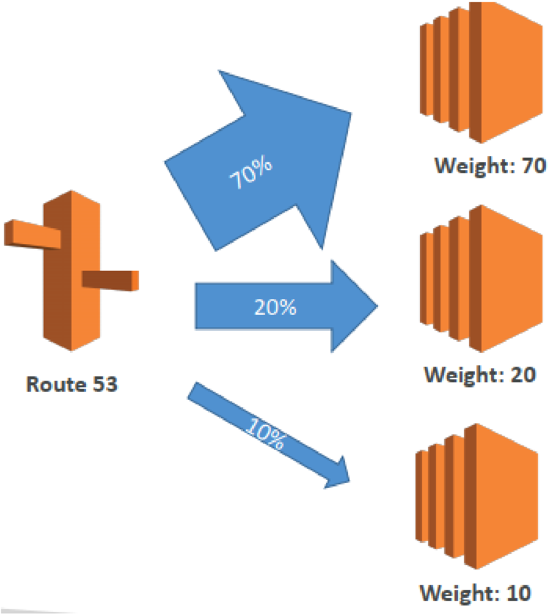
Simple Routing Policy

- Maps a hostname to another hostname
- Use when you need to redirect to a single resource
- You can't attach health checks to simple routing policy
- **If multiple values are returned, a random one is chosen by the client**



Weighted Routing Policy

- Control the % of the requests that go to specific endpoint
- Helpful to test 1% of traffic on new app version for example
- Helpful to split traffic between two regions
- Can be associated with Health Checks



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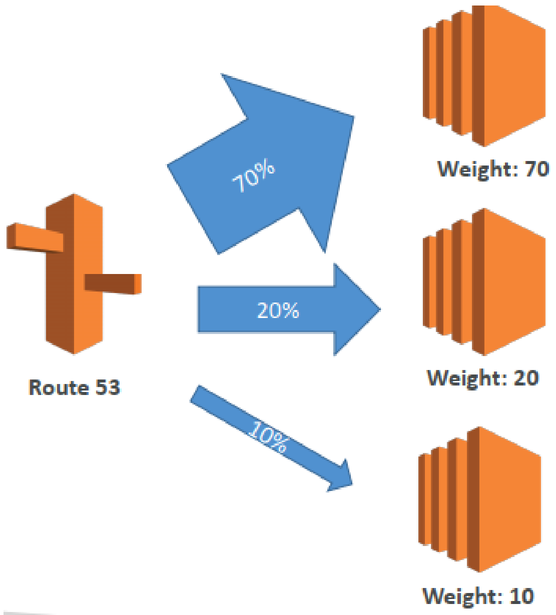


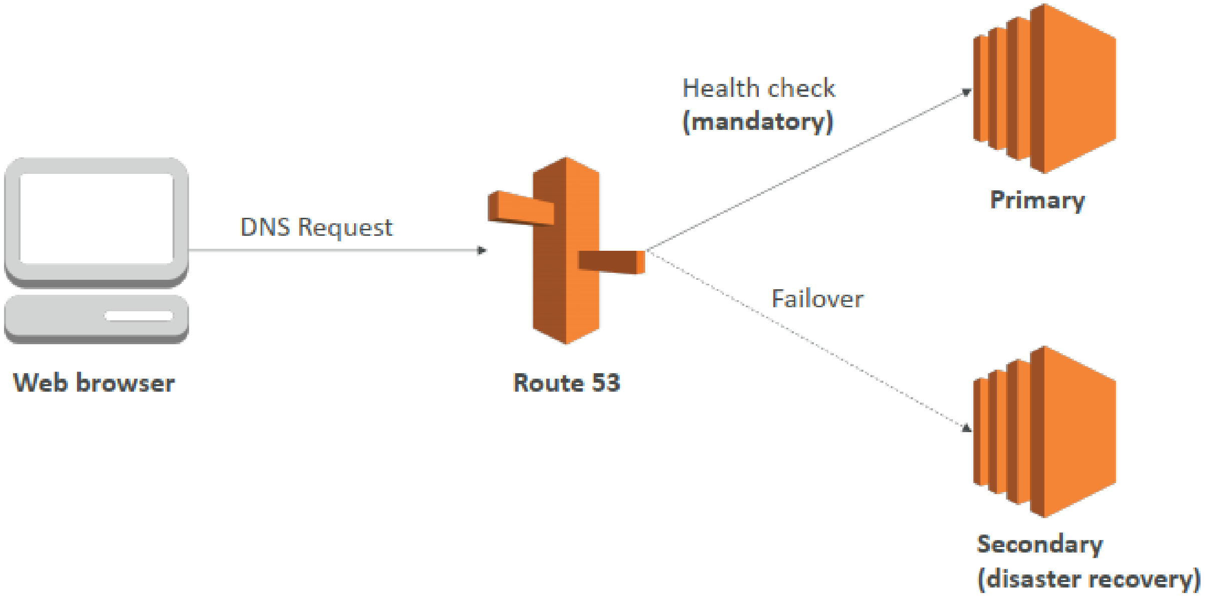
Latency Routing Policy

- Redirect to the server that has the least latency close to us
- Super helpful when latency of users is a priority
- Latency is evaluated in terms of user to designed AWS Region
- Germany may be directed to the US (if that’s the lowest latency)

Health Checks

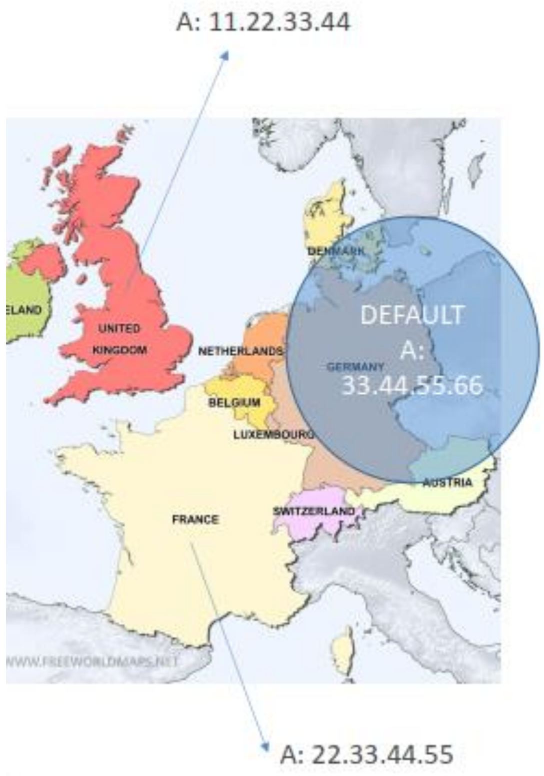
- Have X health checks failed -> unhealthy (default 3)
- After X healthy checks passed -> healthy (default 3)
- Default Healthy Check Interval: 30s (can set to 10s – higher cost)
- About 15 healthy checkers will check the endpoint health
- -> one request every 2 seconds on average
- Can have HTTP, TCP and HTTPS healthy checks (no SSL verification)
- Possibility of integrating the health check with CloudWatch
- Healthy checks can be linked to Route53 DNS queries

Failover Routing Policy



Geo Location Routing Policy

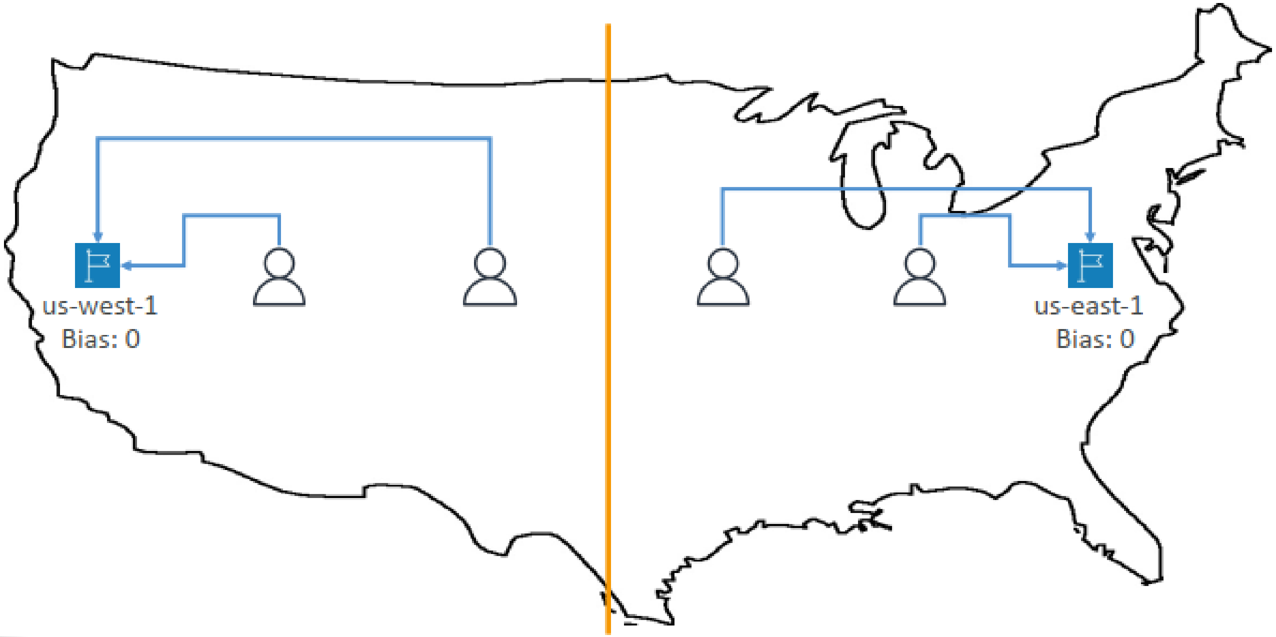
- Different from Latency based
- This is routing based on user location
- Here we specify: traffic from the UK should go to this specific IP
- Should create a "default" policy (in case there's no match on location)



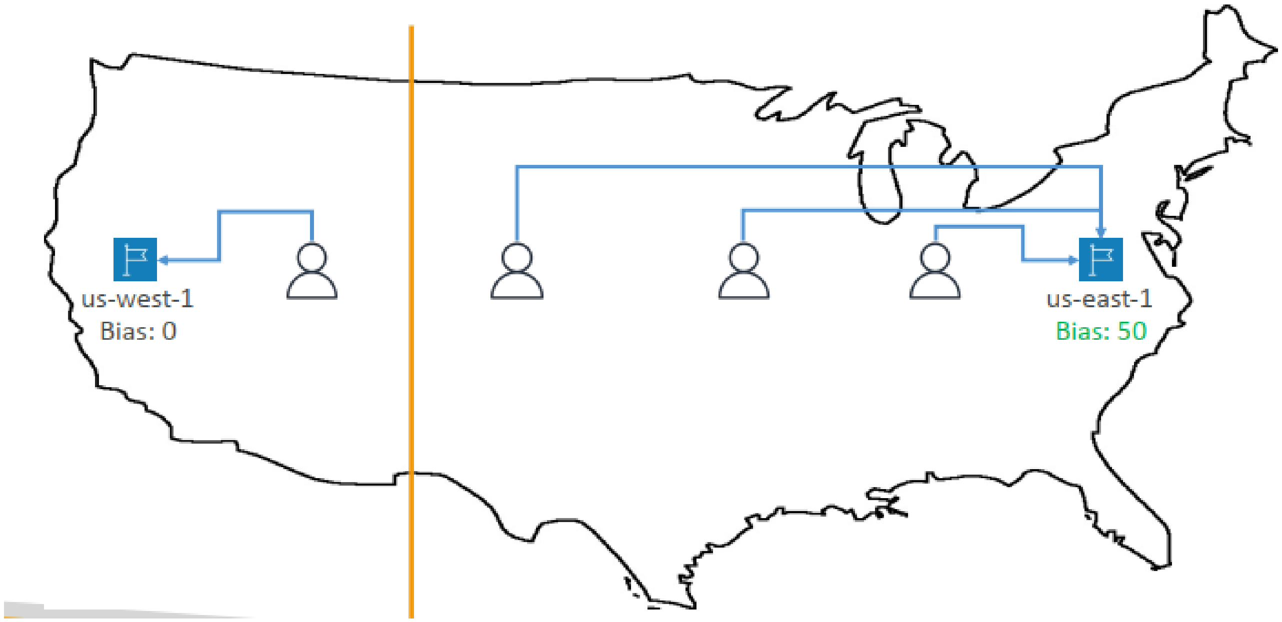
Geoproximity Routing Policy

- Route traffic to your resources based on the geographic location of users and resources
- Ability to shift more traffic to resources based on the defined bias
- To change the size of the geographic region, specific bias values
- Resources can be
  - AWS resources (specify AWS region)
  - Non-AWS resources (specify Latitude and Longitude)
- You must use Route 53 Traffic Flow (advanced) to use this feature

Geoproximity Routing Policy



Geoproximity Routing Policy



Multi Value Routing Policy

- Use when routing traffic to multiple resources
- Want to associate a Route 53 health checks with records
- Up to 8 healthy records are returned for each Multi Value query
- MultiValue is not a substitute for having an ELB

Name	Type	Value	TTL	Set ID	Health Check
www.example.com	A Record	192.0.2.2	60	Web1	A
www.example.com	A Record	198.51.100.2	60	Web2	B
www.example.com	A Record	203.0.113.2	60	Web3	C

Route53 as a Registrar

- A domain name registrar is an organization that manages the reservation of Internet domain names
- Famous names
  - GoDaddy
  - Google Domains
  - Etc…
- And also … Route53 (e.g. AWS)!
- Domain Registrar != DNS

3rd Party Registrar with AWS Route 53

- If you buy your domain on 3rd party website, you can still use Route53
- 1) Create a Hosted Zone in Route 53
- 2) Update NS Records on 3rd party website to use Route 53 name servers

Thank you!!!